

★ 수열의 합 (S_n) 과 일반항 (a_n) 관계

$$a_n = \begin{cases} S_n & (n=1) \\ S_n - S_{n-1} & (n \geq 2) \end{cases}$$

a_1	a_2	a_3	a_4	a_5	a_6	...
1	3	5	7	9	11	...

$$S_1 = a_1$$

$$S_2 = a_1 + a_2$$

$$S_3 = a_1 + a_2 + a_3$$

$$S_4 = a_1 + a_2 + a_3 + a_4$$

#예3-8) $S_n = n^2 + 2n$

$$\begin{aligned} \text{i) } a_n &= S_n - S_{n-1} = (n^2 + 2n) - [(n-1)^2 + 2(n-1)] \\ &= (\cancel{n^2} + 2n) - (\cancel{n^2} - \cancel{2n} + 1 + \cancel{2n} - 2) \\ &= 2n + 1 \quad (n \geq 2) \end{aligned}$$

$$\text{ii) } n=1 \text{ 일 때 } a_n = S_n \text{ 이므로,}$$

$$a_1 = S_1 = 1^2 + 2 = 3$$

$$\therefore \text{i), ii) 에 의해}$$

$$a_n = 2n + 1 \text{ 이다.}$$

점화식

예1) $a_{n+1} = a_n + (n+1)$, $a_{n+1} = a_n + (4n^2 + 7)$

① $a_{n+1} = a_n + b_n \Rightarrow a_n = a_1 + \sum_{k=1}^{n-1} b_k$

$$a_2 = a_1 + b_1$$

$$a_3 = a_2 + b_2$$

$$a_4 = a_3 + b_3$$

⋮

$$+ \left| a_n = a_{n-1} + b_{n-1} \right.$$

$$a_2 + a_3 + a_4 + \dots + a_n = (a_1 + b_1) + (a_2 + b_2) + (a_3 + b_3) + \dots + (a_{n-1} + b_{n-1})$$

$$a_n = a_1 + b_1 + b_2 + b_3 + \dots + b_{n-1} = a_1 + \sum_{k=1}^{n-1} b_k$$

예2-9. $a_1 = 3$ $a_{n+1} = a_n - 3n^2$

$$a_n = a_1 + \sum_{k=1}^{n-1} b_k = 3 + \sum_{k=1}^{n-1} (-3k^2) = 3 - 3 \sum_{k=1}^{n-1} k^2$$

$$= 3 - 3 \cdot \frac{(n-1)n(2n-1)}{2}$$

$$= \frac{n(2n^2 - 3n + 1)}{2} = \frac{-(n(n-1)(2n-1)) - 6}{2} = -\frac{2n^3 - 3n^2 + n - 6}{2}$$

$$\sum_{k=1}^{n-1} k^2 = \frac{(n-1)n(2n-1)}{6} \quad \begin{matrix} \text{red} \\ 2(n-1)+1 \\ = 2n-1 \end{matrix}$$

$$\text{ex 1) } a_{n+1} = (n+1)a_n, \quad a_{n+1} = \frac{1}{n+1} a_n$$

$$\textcircled{2} \quad a_{n+1} = b_n a_n \Rightarrow a_n = a_1 \cdot b_1 b_2 b_3 \cdots b_{n-1}$$

$$a_2 = b_1 a_1$$

$$a_3 = b_2 a_2$$

$$a_4 = b_3 a_3$$

⋮

$$\times \left\{ \begin{array}{l} a_n = b_{n-1} a_{n-1} \end{array} \right.$$

$$a_2 a_3 a_4 \cdots a_n = b_1 b_2 b_3 b_4 \cdots b_{n-1} a_1 a_2 a_3 a_4 \cdots a_{n-1}$$

$$a_n = b_1 b_2 b_3 b_4 \cdots b_{n-1} \cdot a_1$$

$$\text{\# ex 3-10. } a_1 = 1, \quad a_{n+1} = \frac{n}{n+1} a_n$$

$$a_n = a_1 \cdot b_1 b_2 b_3 \cdots b_{n-1}$$

$$= \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{4}{5} \cdots \frac{n-1}{n} = \frac{1}{n}$$

$$\therefore a_n = \frac{1}{n}$$

③ $a_{n+1} = ca_n + d \Rightarrow$ 등비수열로 푼다!

$$\underbrace{a_{n+1} - d}_{b_{n+1}} = c \underbrace{(a_n - d)}_{b_n} \quad \begin{array}{l} \rightarrow a_{n+1} - d = c(a_n - d) \\ a_{n+1} = ca_n - \underbrace{cd + d}_d \end{array}$$

$b_n = a_n - d$
 $b_{n+1} = a_{n+1} - d$

$$b_{n+1} = c b_n$$

$$b_n = b_1 \cdot c^{n-1}$$

$$a_n - d = (a_1 - d) \cdot c^{n-1}$$

#예3-1)

$$a_1 = 3, \quad a_{n+1} = 4a_n - 3$$

$$\underbrace{a_{n+1} - d}_{b_{n+1}} = 4 \underbrace{(a_n - d)}_{b_n} \Rightarrow \begin{array}{l} a_{n+1} = 4a_n - 4d + d \\ = 4a_n - 3d = 4a_n - 3 \end{array} \quad \therefore d = 1$$

$$b_{n+1} = 4 b_n$$

$$b_n = b_1 \cdot 4^{n-1}$$

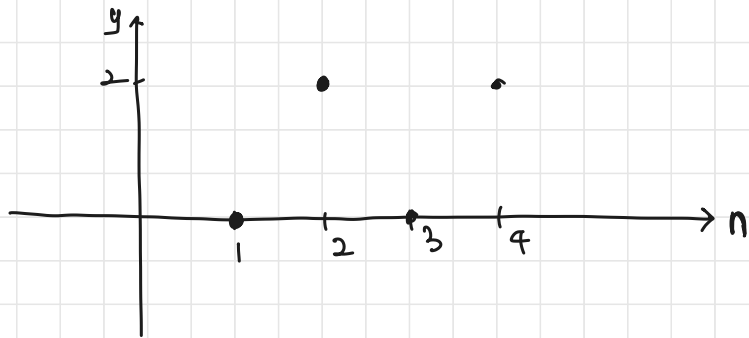
$$a_n - d = (a_1 - d) \cdot 4^{n-1}$$

$$a_n - 1 = (3 - 1) \cdot 4^{n-1} \Leftrightarrow a_n = 2 \cdot 4^{n-1} + 1$$

함수의 극한

$$\lim_{x \rightarrow \infty} \frac{1}{x} = \frac{1}{\infty} = 0$$

$$\lim_{x \rightarrow \infty} \frac{11x^3 + 3}{4x^2} = \infty$$

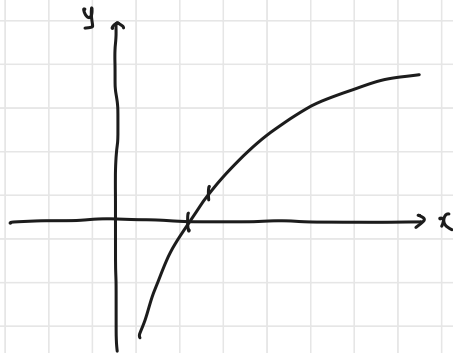


수열의 극한.

$$\lim_{n \rightarrow \infty} \frac{1}{n} = \lim_{n \rightarrow \infty} a_n$$

$$\lim_{n \rightarrow \infty} \frac{11n^3 + 3}{4n^2} = ?$$

발산 $\rightarrow$ 진동.

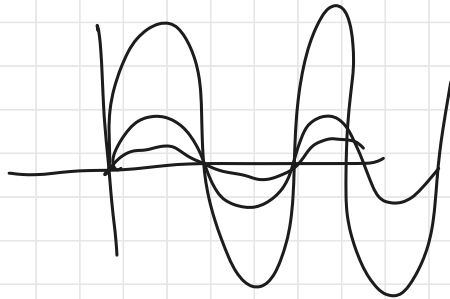


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$$(a+b)(a-b) = a^2 - b^2$$

$$\lim_{n \rightarrow \infty} (\sqrt{n^2 - 2n} - n) = \lim_{n \rightarrow \infty} \frac{(\sqrt{n^2 - 2n} - n)(\sqrt{n^2 - 2n} + n)}{\sqrt{n^2 - 2n} + n}$$

$$= \lim_{n \rightarrow \infty} \frac{(n^2 - 2n) - n^2}{\sqrt{n^2 - 2n} + n} = \lim_{n \rightarrow \infty} \frac{-2n}{\sqrt{n^2 - 2n} + n} = \lim_{n \rightarrow \infty} \frac{-2}{\sqrt{\frac{n^2 - 2n}{n^2}} + 1} = \lim_{n \rightarrow \infty} \frac{-2}{\sqrt{1 - \frac{2}{n}} + 1} = \frac{-2}{2} = -1$$



$$\lim_{x \rightarrow \infty} \frac{-1}{x} \leq \lim_{x \rightarrow \infty} \frac{\sin x}{x} \leq \lim_{x \rightarrow \infty} \frac{1}{x}$$

$$0 \leq \lim_{x \rightarrow \infty} \frac{\sin x}{x} \leq 0$$